

Erdem Bıyık

<https://ebiyik.github.io>

<https://scholar.google.com/citations?user=P-G3sjYAAAAJa>

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Room 305C, Los Angeles, CA 90089

CURRENT POSITION

- | | |
|---|---------------------------|
| University of Southern California | Los Angeles, CA, USA |
| • <i>Assistant Professor of Computer Science</i> | <i>Aug 2023 – present</i> |
| <i>Assistant Professor of Electrical and Computer Engineering - Systems (by courtesy)</i> | <i>Jan 2024 – present</i> |

EDUCATION

- | | |
|--|----------------------------|
| Stanford University | Stanford, CA, USA |
| • <i>Doctor of Philosophy in Electrical Engineering; GPA: 4.07/4.00; Advisor: Prof. Dorsa Sadigh</i> | <i>Sep 2017 – Jun 2022</i> |
| Stanford University | Stanford, CA, USA |
| • <i>Master of Science in Electrical Engineering; GPA: 4.08/4.00</i> | <i>Sep 2017 – Apr 2019</i> |
| Bilkent University | Ankara, Türkiye |
| • <i>Bachelor of Science in Electrical and Electronics Engineering; GPA: 4.00/4.00; Rank: 1/170</i> | <i>Aug 2012 – Jun 2017</i> |
| National University of Singapore | Singapore |
| • <i>Exchange Student in Electrical and Computer Engineering</i> | <i>Aug 2015 – Dec 2015</i> |

WORK EXPERIENCE

- | | |
|--|---|
| UC Berkeley, Center for Human-Compatible Artificial Intelligence (CHAI) | Berkeley, CA, USA |
| • <i>Postdoctoral Researcher</i> | <i>Jul 2022 - Jul 2023</i> |
| ◦ Reward Learning: Working on various active reward learning projects for robotics with Prof. Stuart Russell and Prof. Anca Dragan | |
| ◦ Reinforcement Learning for Deep Brain Stimulation: Working on the applications of reinforcement learning algorithms for deep brain stimulation under the supervision of Prof. Anca Dragan | |
| Google Research | Mountain View, CA, USA |
| • <i>Research Intern</i> | <i>Jun 2021 - Sep 2021</i> |
| ◦ Recommender Systems: Working on active preference elicitation for recommender systems under the supervision of Dr. Yinlam Chow, Dr. Mohammad Ghavamzadeh, and Prof. Craig Boutilier | |
| ◦ Preference-based Reinforcement Learning: Working on relaxing initial state assumptions in preference-based reinforcement learning under the supervision of Dr. Yinlam Chow and Dr. Mohammad Ghavamzadeh | |
| National Magnetic Resonance Research Center (UMRAM) | Ankara, Türkiye |
| • <i>Undergraduate Researcher & Research Intern</i> | <i>Apr 2016 - Sep 2017</i> |
| ◦ csMRI: Developing compressed sensing methods for accelerated MRI under the supervision of Prof. Tolga Çukur | |
| ◦ SSFP Imaging: Developing coil compression methods and artifact suppression techniques for balanced SSFP under the supervision of Prof. Tolga Çukur | |
| ASELSAN | Ankara, Türkiye |
| • <i>Intern (2015), Research Engineer (2017)</i> | <i>Jun 2015 - Jul 2015, Apr 2017 - Aug 2017</i> |
| ◦ Military Communications (2015): Developing a C++ program to decode and encode data in MIL-STD-3014 protocol with an easy-to-use interface | |
| ◦ Algorithms Design (2015): Designing algorithms to solve composite launch acceptability region problem | |
| ◦ SSFP Imaging (2017): R&D projects related to field inhomogeneity, banding profile estimation, coil compression and banding suppression in bSSFP MRI under the supervision of Dr. Aykut Koç | |
| École polytechnique fédérale de Lausanne (EPFL) | Lausanne, Switzerland |
| • <i>Research Intern via Summer@EPFL Program</i> | <i>Jun 2016 - Sep 2016</i> |
| ◦ Approximate Message Passing: Research about approximate message passing algorithms as an intern in the Information Processing Group under the supervision of Prof. Rüdiger Urbanke | |
| Anadolu Agency | Ankara, Türkiye |
| • <i>Intern</i> | <i>Jun 2014 - Jul 2014</i> |

- **Image Processing:** Developing a Java, C++ and MySQL based program that uses image registration, matching and comparison techniques to detect copyright infringements

TEACHING EXPERIENCE

- **University of Southern California** Los Angeles, CA
Instructor *Fall 2023, Spring 2024, Fall 2024, Fall 2025*
 - **CSCI445L Introduction to Robotics (Spring 2024):** Teaching the course
 - **CSCI699 Robot Learning (Fall 2023, Fall 2024, Fall 2025):** Designing and teaching the course
- **Stanford University** Stanford, CA
Teaching Assistant *Winter 2020, Winter 2021*
 - **CS237B / EE260B / AA174B / AA274B Principles of Robot Autonomy II:** Holding office hours and sections, preparing and grading homeworks and exams
Awarded “outstanding course assistant” by the CS department (top 5%) in Winter 2021
- **Bilkent University** Ankara, Türkiye
Teaching Assistant *Spring 2014, Fall 2016*
 - **CS114 Introduction to Programming for Engineers (2014):** Teaching in the weekly tutorials & recitations
 - **EEE211 Analog Electronics (2016):** Teaching, interviewing and grading students in the laboratory sessions

MENTORING

- **Current Ph.D. Students:** Anthony Liang (co-advised with Stephen Tu), Pavel Czempin, Yigit Korkmaz, Yutai Zhou, Yimin Tang (co-advised with Sven Koenig), Ayano Hiranaka (co-advised with Daniel Seita), Bo-Ruei Huang
- **Current Master’s Students:** Xinhui Li, Dhanush Kumar Varma Penmetsa, Matthew Hong, Yusen Luo, Zhaoyang Li, Litian Gong, Abha Jha, Cagan Bakirci, Asmaa Hassan, Abhinav Srivatsa
- **Current Undergraduate Students:** Jaiv Doshi, Urvi Bhuwania, Ashley Lan, Kevin Kim, Arpita Tomar, Yasir White, Qingyang (Bob) Zeng, Alina Du, Jingzhen (Steve) Wang, Meredith Li, Fatemeh Bahrani
- **Ph.D. Alumni**
 - Jesse Zhang (2025, co-advised with Jesse Thomason and Joseph Lim). Next: Postdoc at UW.
 - Ayush Jain (2024, co-advised with Joseph Lim). Next: Research Scientist at Meta.
 - Sumedh Sontakke (2024, co-advised with Laurent Itti). Next: Applied Scientist at Amazon Robotics.
- **Master’s Alumni**
 - Mihir Kulkarni (2025). Next: Ph.D. Student in Computer Science at UT Dallas.
 - Zhaojing Yang (2024). Next: Ph.D. Student in Computer Science at UC San Diego.
 - Jiahui Zhang (2024). Next: Ph.D. Student in Computer Science at UT Dallas.
 - Thomas Reeves (2024). Next: Ph.D. Student in Computer Science at USC.
- **Undergraduate Alumni**
 - Shreyas Ramanujam (2025, Intern from IIT Madras). Next: Master’s Student at Stanford University.
 - Eisuke Hirota (2025, Intern from SUNY Binghamton). Next: MIT Lincoln Laboratory.
 - Miru Jun (2024). Next: Data Engineer at US Navy.

HONORS & AWARDS

- **RLC 2025 Outstanding Paper Award on Empirical Reinforcement Learning Research:** Awarded for the paper “Mitigating Suboptimality of Deterministic Policy Gradients in Complex Q-functions”
- **Best Paper Award at the “2nd Workshop on Out-of-Distribution Generalization in Robotics: Towards Reliable Learning-Based Autonomy” at RSS 2025:** Awarded to the top paper among ~50 papers. Awarded for the paper “ReWiND: Language-Guided Rewards Teach Robot Policies without New Demonstrations”
- **2024 Outstanding Paper Finalist at TMLR:** Awarded to 5 papers in 2024. Awarded for the paper “Open Problems and Fundamental Limitations of Reinforcement Learning from Human Feedback”
- **Stanford CS Outstanding Course Assistant:** Awarded to the top 5% of CAs in the department. Awarded for the “Principles of Robot Autonomy II” course in Winter 2021
- **Qualcomm Innovation Fellowship 2020 North America:** Finalist (as a group of two PhD students)

- **HRI 2020 Honorable Mention in Technical Advances:** Awarded to 3 papers in the Technical Advances track. Awarded for the paper “When Humans Aren’t Optimal: Robots that Collaborate with Risk-Aware Humans”
- **Qualcomm Innovation Fellowship 2019 North America:** Finalist (as a group of two PhD students)
- **Stanford University James D. Plummer Graduate Fellowship (2017-2018):** Full tuition waiver & stipend during the first year of PhD program
- **Bilkent University Electrical and Electronics Engineering Department, Graduation Awards (2017):** Academic Excellence, Research Excellence, Voluntary Professional Activities, Social Awareness and Activities
- **Scholarship of the Turkish Prime Ministry (2012-2017):** Awarded monthly stipend during the BSc program (given to those who rank in first 100 among 1.8 million students in nationwide university entrance exam)
- **Bilkent University Comprehensive Scholarship (2012-2017):** Full tuition waiver & stipend during the BSc program
- **IEEEExtreme Programming Competitions:** 2nd in Türkiye, 78th among all participants, 2015. 1st in Türkiye, 73rd among all participants, 2014. 1st in Türkiye, 100th among all participants, 2013 (All as a group of three students)
- **Turkish Intelligence Foundation (TZV) Marathon:** Ranked twice in top 25 and once 6th, 2012-2014
- **İşbank Golden Youth Award (2012):** Granted for outstanding performance in the nationwide university entrance exam
- **Nationwide University Entrance Exam (LYS):** Ranked 13th among 1.8 million students in Türkiye, 2012

INVITED TALKS & POSTER PRESENTATIONS

- Maximally Informative, Minimally Demanding: Learning from Human Feedback
 - UCLA, Computer Science, Department Seminar (2025)
 - Carnegie Mellon University, Guest Lecture at “Human-Robot Interaction” (2025)
 - Inha University, Department of Aerospace Engineering (2025)
 - CoRL 2025 Workshop on Resource-Rational Robot Learning
- SPUR: Scaling Reward Learning from Human Demonstrations
 - CoRL 2025 Workshop on Evaluation and Deployment Across the Robot Learning Lifecycle (Eval&Deploy)
- Robots That Pay Attention, Follow Instructions, and Learn Stably
 - Sony AI (2025)
- Robot Learning with Minimal Human Feedback
 - Advancing AI Capacity in Indonesian Universities: A Collaborative Workshop for Faculty Members (2025)
 - Stanford University, Guest Lecture at “Principles of Robot Autonomy II” (2025)
 - University of Pennsylvania, General Robotics, Automation, Sensing, & Perception Lab (GRASP) Seminar (2025)
 - UC Santa Barbara, The Center for Control, Dynamical Systems, and Computation Seminar (2025)
 - Bilkent University, Electrical and Electronics Engineering Department, Graduate Seminar (2024)
 - University of Washington, Robotics Seminar (2024)
 - University of Illinois Urbana-Champaign, Human-Centered Autonomy Lab (2024)
- Learning Humans’ Tradeoffs between Objectives via Language Feedback
 - RSS 2025 Workshop on Multi-Objective Optimization and Planning in Robotics
- Multi-Agent Inverse Q-Learning from Demonstrations
 - ICRA 2025
- RAILGUN: A Unified Convolutional Policy for Multi-Agent Path Finding Across Different Environments and Tasks
 - ICRA 2025 Workshop on Foundation Models and Neuro-Symbolic AI for Robotics (poster)
- Asking Easy Questions: A User-Friendly Approach to Active Reward Learning
 - Inzva & AI Safety Türkiye, AI Safety Talks (2025)
 - CoRL 2019 (poster)
- Open Problems in Reinforcement Learning from Human Feedback and Potential Solutions for Data-Efficiency
 - Caltech AI Alignment Club (2025)
 - University of Amsterdam, The Amsterdam Machine Learning Lab (2023)
- Efficient Robot Learning via Interaction with Humans
 - New Faculty Highlights at the 39th Annual AAAI Conference on Artificial Intelligence, 2025
 - Southern California Robotics (SCR) Symposium 2023

- Reinforcement Learning from Comparative Language Feedback
 - The Inaugural Conference of the International Association for Safe and Ethical AI (IASEAI 2025 – poster)
- Trajectory Improvement and Reward Learning from Comparative Language Feedback
 - CoRL 2024 (poster)
- Preference Learning from Minimal Human Feedback for Interactive Autonomy
 - ODSC West 2024 Conference (Open Data Science Conference)
 - International Conference on Micro, Nano and Smart Systems (IC-MNSS) 2024
 - IIT Hyderabad, Mechanical and Aerospace Engineering, Department Seminar & Technology Innovation Hub on Autonomous Navigation and Data Acquisition Systems (2024)
- Making Robot Learning More Natural Through Human Saliency and Language
 - RSS 2024 Workshop on “Mechanisms for Mapping Human Input to Robots: From Robot Learning to Shared Control/Autonomy”
- Data- and Time-Efficient Learning of Human Preferences
 - 8th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop (2024)
- Optimizing Robot Behavior via Comparative Language Feedback
 - Human-Interactive Robot Learning (Workshop at HRI 2024 – poster)
- Data-Efficient Learning from Human Feedback and Large Pretrained Models for Robotics
 - Arizona State University, School of Computing and Augmented Intelligence (2023)
- ViSaRL: Visual Reinforcement Learning Guided by Human Saliency
 - 7th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop (2023 – poster)
- Learning Preferences for Interactive Autonomy
 - Middle East Technical University, Robotics and AI Technologies Application and Research Center (2022)
 - Sonoma State University, Engineering Colloquium (2022)
 - Cornell University, Robotics Seminar (2022)
 - University of Wisconsin-Madison, Computer Science, Department Seminar (2022)
 - University of Illinois Urbana-Champaign, Computer Science, Department Seminar (2022)
 - University of Michigan, Robotics Institute, Department Seminar (2022)
 - University of Southern California, Computer Science, Department Seminar (2022)
 - Imperial College London, Department of Aeronautics, Aerodynamics & Control Seminar (2022)
 - Carnegie Mellon University, Robotics Institute, Department Seminar (2022)
 - UCLA, Electrical and Computer Engineering, Department Seminar (2022)
 - Bilkent University, Electrical and Electronics Engineering, Graduate Seminar (2021)
 - UC Berkeley, Center for Human-Compatible Artificial Intelligence (CHAI), Beneficial AI Seminar (2021)
 - Sabancı University, Computer Science & Engineering, Department Seminar (2021)
 - Koç University, AI Meetings (2021)
 - Bay Area Robotics Symposium (BARS) 2021 (poster)
 - Virginia Tech, Mechanical Engineering, Department Seminar (2021)
 - Caltech Yue Lab (2021)
 - UT Austin, Personal Autonomous Robotics Lab (2021)
 - UC Berkeley, InterACT Lab (2021)
- Learning Multimodal Rewards from Rankings
 - 6th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop (2022)
 - CoRL 2021
- Learning from Human Preferences
 - Stanford University, Guest Lecture at “Safe and Interactive Robotics” (2022)
- Behavior Modeling, via Reward Learning and for Autonomous Driving
 - Nuro (2022)
- Learning from Humans for Adaptive Interaction
 - HRI Pioneers 2022
- APReL: A Library for Active Preference-based Reward Learning Algorithms

- Human-Interactive Robot Learning (HIRL) (Workshop at HRI 2022)
 - HRI 2022
 - AI-HRI 2021 at AAAI Fall Symposium Series
- Partner-Aware Algorithms in Decentralized Cooperative Bandit Teams
 - AAAI 2022
 - AI-HRI 2021 at AAAI Fall Symposium Series
- The Role of Representations in Human-Aware Learning and Control (*with Dorsa Sadigh*)
 - “Aware-Learning: How to Benefit from Priors” Workshop @ CDC 2021
- Interactive Robotics through the Lens of Learning (*with Dorsa Sadigh*)
 - NCCR (The National Centre of Competence in Research, Switzerland) Automation Seminar, 2021
- Leveraging Smooth Attention Prior for Multi-Agent Trajectory Prediction
 - Center for Automotive Research at Stanford (CARS) Annual Meeting (2021 – poster)
- Learning Reward Functions from Scale Feedback
 - CoRL 2021 (poster)
- Learning how to Dynamically Route Autonomous Vehicles on Shared Roads
 - ETH Zurich, Institute for Dynamic Systems and Control, Autonomy Talks (2021)
 - The 32nd IEEE Intelligent Vehicles Symposium Workshop (2021)
 - 3rd NorCal Control Workshop (2021)
 - Stanford Robotics Lunch (2019)
 - Center for Automotive Research at Stanford (CARS) Annual Meeting (2020 – poster)
- Walking the Boundary of Learning and Interaction (*with Dorsa Sadigh*)
 - 3rd Robot Learning Workshop: Grounding Machine Learning Development in the Real World @ NeurIPS 2020
- When Humans Aren’t Optimal: Robots that Collaborate with Risk-Aware Humans (*with Minae Kwon*)
 - University of Chicago, Graduate Seminar on Topics in Human-Robot Interaction (2020)
- Active Preference-Based Gaussian Process Regression for Reward Learning
 - RSS 2020
- The Green Choice: Learning and Influencing Human Decisions on Shared Roads
 - CDC 2019
- Active Learning of Reward Dynamics from Hierarchical Queries
 - IROS 2019
- Efficient and Safe Exploration in Deterministic Markov Decision Processes with Unknown Transition Models
 - ACC 2019
 - Stanford AI Safety Retreat 2019 (poster)
- Batch Active Preference-Based Learning of Reward Functions
 - CoRL 2018
 - Stanford HAI 2019 (poster)
 - TRI Joint University Workshop 2019 (poster)
 - Stanford SystemX Fall 2018 (poster)
 - BARS - Bay Area Robotics Symposium (BARS) 2018 (poster)

PROFESSIONAL ACTIVITIES

- **Conferences Organized**
 - **Local Arrangements Co-Chair**, Robotics: Science and Systems (RSS), 2025
 - **Organizer**, Human-in-the-Loop Robot Learning: Teaching, Correcting, and Adapting (Workshop at Robotics: Science and Systems (RSS)), 2025
 - **Finance Chair**, Conference on Robot Learning (CoRL), 2024
 - **Local Arrangements Chair**, Bay Area Robotics Symposium (BARS), 2021
 - **Local Arrangements Chair**, Bay Area Robotics Symposium (BARS), 2020

- **Organizer**, Emergent Behaviors in Human-Robot Systems (Workshop at Robotics: Science and Systems (RSS)), 2020
- **Local Arrangements Chair**, Bay Area Robotics Symposium (BARS), 2018
- **Program Committee Member**
 - AAAI 2026 New Faculty Highlights Track
 - ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2025
 - The 39th Annual AAAI Conference on Artificial Intelligence, AI Alignment Track (2025)
 - 8th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop, 2024
 - 7th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop, 2023
- **Editorial Board**
 - **Associate Editor** for Robot Learning, IEEE Robotics and Automation Letters (RA-L), 2023 – present
 - **Associate Editor**, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025
 - **Area Chair** for Robotics: Science and Systems - Pioneers Workshop 2025
 - **Area Chair** for Late Breaking Results at the ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2025
 - **Associate Editor**, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024
 - **Area Chair**, Responsible AI (RAI) (Workshop at International Conference on Learning Representations (ICLR)), 2021
- **Panelist**
 - RSS 2025 Workshop on Multi-Objective Optimization and Planning in Robotics
 - Human Value Learning Panel at the 8th Annual Center for Human-Compatible Artificial Intelligence (CHAI) Workshop, 2024
- **Reviewer for Journals**
 - ACM Transactions on Autonomous and Adaptive Systems (TAAS)
 - ACM Transactions on Human-Robot Interaction (THRI)
 - ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)
 - Artificial Intelligence
 - Autonomous Agents and Multi-Agent Systems
 - Autonomous Robots (AURO)
 - Cognitive Systems Research
 - Frontiers in Computer Vision
 - Frontiers in Robotics and AI
 - Nature Communications
 - IEEE Control System Letters (L-CSS)
 - IEEE Robotics and Automation Letters (RA-L)
 - IEEE Robotics and Automation Magazine (RAM)
 - IEEE Transactions on Artificial Intelligence (TAI)
 - IEEE Transactions on Automatic Control (TAC)
 - IEEE Transactions on Automation Science and Engineering (T-ASE)
 - IEEE Transactions on Cognitive and Developmental Systems (TCDS)
 - IEEE Transactions on Control Systems Technology (TCST)
 - IEEE Transactions on Human-Machine Systems (THMS)
 - IEEE Transactions on Intelligent Transportation Systems (T-ITS)
 - IEEE Transactions on Knowledge and Data Engineering (TKDE)
 - IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
 - IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
 - IEEE Transactions on Robotics (T-RO)
 - IEEE Transactions on Systems, Man and Cybernetics: Systems (TSMC-S)
 - International Journal of Machine Learning and Cybernetics
 - International Journal of Robotics Research (IJRR)

- Journal of Machine Learning Research (JMLR)
- MDPI Robotics
- Proceedings of the Royal Society A
- Robotica
- **Reviewer for Conferences**
 - International Conference on Learning Representations (ICLR), 2025-2026
 - IEEE International Conference on Robotics and Automation (ICRA), 2021–2024, 2026
 - Conference on Robot Learning (CoRL), 2019–2025
 - IEEE Conference on Decision and Control (CDC), 2019–2020, 2023–2025
 - Robotics: Science and Systems (RSS), 2020–2021, 2023–2025
 - Conference on Uncertainty in Artificial Intelligence (UAI), 2023–2025 (*Top Reviewer Recognition in 2025*)
 - International Conference on Machine Learning (ICML), 2025
 - International Conference on Artificial Intelligence and Statistics (AISTATS), 2025
 - Conference on Neural Information Processing Systems (NeurIPS), 2023–2024
 - IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020, 2022–2024
 - IEEE International Conference on Robot & Human Interactive Communication (RO-MAN), 2024
 - Learning for Dynamics & Control (L4DC), 2020, 2023–2024
 - ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2020–2024
 - International Joint Conference on Artificial Intelligence (IJCAI), 2023
 - American Control Conference (ACC), 2019, 2021–2022
 - IEEE Conference on Control Technology and Applications (CCTA), 2021
 - IEEE-RAS International Conference on Humanoid Robots (Humanoids), 2020
 - International Conference on Computer-Aided Verification (CAV), 2019
 - ACM International Conference on Hybrid Systems: Computation and Control (HSCC), 2019
- **Reviewer for Workshops**
 - 2nd Workshop on Semantic Reasoning and Goal Understanding in Robotics (SemRob) at RSS 2025
 - Workshop on User-Aligned Assessment of Adaptive AI Systems (AIA) at IJCAI 2025
 - RSS Pioneers Workshop 2023–2025
 - HRI Pioneers Workshop 2023–2024
 - Interactive Learning with Implicit Human Feedback Workshop (ILHF) at ICML 2023
 - AAMAS Workshop on Rebellious and Disobedient AI (RaD-AI), 2022–2023
 - CoRL Workshop on Aligning Robot Representations with Humans, 2022
 - NeurIPS Workshop on Foundation Models for Decision Making (FMDM), 2022
 - NeurIPS Workshop on Progress and Challenges in Building Trustworthy Embodied AI (TEA), 2022
 - NeurIPS Workshop on Machine Learning Safety, 2022
 - RSS Workshop on Learning from Diverse, Offline Data (L-DOD), 2022
 - RSS Workshop on Social Intelligence in Humans and Robots, 2022
 - Artificial Intelligence for Human-Robot Interaction Symposium (AI-HRI) at AAAI Fall Symposium Series, 2021
 - ICRA Workshop on Social Intelligence in Humans and Robots, 2021
 - Bridging AI and Cognitive Science (BAICS) (Workshop at International Conference on Learning Representations (ICLR)), 2020
- **Session Chair at Conferences**
 - IROS 2025
 - RSS 2024
 - ICMNSS 2024
 - ICRA 2024

COMMITTEE MEMBERSHIP FOR OTHER STUDENTS

- **University of Southern California, Computer Science**
 - Grace Zhang, Qualifying Exam (May 2024), Thesis Proposal (Apr. 2025), Defense (Nov. 2025)

- Sumeet Batra, Defense (Nov. 2025)
- Varun Sreedhara Bhatt, Qualifying Exam (Feb. 2024), Thesis Proposal (Oct. 2025)
- Zhonghao Shi, Qualifying Exam (Jun. 2024), Thesis Proposal (Jun. 2025)
- Bingjie Tang, Thesis Proposal (May 2024), Defense (May 2025)
- Kegan Strawn, Qualifying Exam (Oct. 2024), Thesis Proposal (May 2025)
- Bryon Tjanaka, Thesis Proposal (Apr. 2025)
- Shihan Zhao, Qualifying Exam (Apr. 2025)
- Soumyaroop Nandi, Thesis Proposal (Apr. 2025)
- Robby Costales, Qualifying Exam (Sep. 2024), Thesis Proposal (Apr. 2025)
- I-Chun Liu, Qualifying Exam (Mar. 2025)
- Anand Balakrishnan, Thesis Proposal (Dec. 2024)
- Sophie Hsu, Qualifying Exam (Dec. 2024)
- Kiran Lekkala, Defense (Nov. 2024)
- Xin Zhu, Thesis Proposal (Jun. 2024)
- Sheryl Paul, Qualifying Exam (May 2024)
- Yang Chen, Qualifying Exam (Mar. 2024)
- Weizhe Chen, Qualifying Exam (Mar. 2024)
- Jeremy Morgan, Qualifying Exam (Nov. 2023)
- **University of Southern California, Electrical and Computer Engineering**
 - Eray Erturk, Qualifying Exam (Nov. 2025)
 - Mingye Li, Qualifying Exam (Jan. 2025), Defense (Aug. 2025)
 - Javier Borquez, Defense (Jun. 2025)
 - Yuning Zhang, Qualifying Exam (May 2025)
 - Zeming Cheng, Qualifying Exam (Sep. 2024), Defense (Apr. 2025)
 - Mustafa Altay Karamuftuoglu, Qualifying Exam (May 2024), Defense (Apr. 2025)

OUTREACH AND MENTORING

- **USC Thomas Lord Department of Computer Science, 2024-present** Los Angeles, CA, USA
 - **Undergraduate Mentoring Program Organizer:** Organizing the program where PhD students mentor undergraduates about their future career, getting involved in CS research, and applying for industry or graduate school
- **USC Robotics and Autonomous Systems Center (RASC), 2024-present** Los Angeles, CA, USA
 - **Faculty Mentor for USC Robotics Seminar (URoS):** Mentoring Ph.D. students in running a biweekly robotics seminar to host students presentations and invited speakers
 - **Faculty Mentor for Robotics and Autonomous Systems Center (RASC) Blog:** Mentoring Ph.D. students in running a blog website about the robotics and autonomous systems research at USC
- **USC Viterbi School of Engineering, K-12 STEM Center, 2025** Los Angeles, CA, USA
 - **Instructor at CS@SC Coding Camps:** Teaching middle school and high school students “how robots learn to imitate humans” in two sessions
- **Berkeley Artificial Intelligence Safety Initiative for Students, 2023-2024** Berkeley, CA, USA
 - **Mentor at Supervised Program for Alignment Research:** Mentoring three undergraduate students on a multi-agent inverse reinforcement learning research project
- **Berkeley Artificial Intelligence Research Lab (BAIR), 2022-2023** Berkeley, CA, USA
 - **BAIR Undergraduate Mentoring Program Mentor:** Mentoring undergraduate students about their future career, getting involved in AI research, and applying for industry or graduate school
- **Stanford University Department of Computer Science, 2020-2022** Stanford, CA, USA
 - **CS Mentorship Program Organizer:** Organizing the CS mentorship program and the related social events
 - **CS Mentorship Program Mentor:** Mentoring first and second-year undergraduate students about their future career, getting involved in research projects, and applying for industry or graduate school
- **Stanford University AI Laboratory (SAIL), 2018-2020** Stanford, CA, USA
 - **AI Mentorship Program Organizer:** Organizing the AI mentorship program and the related social events
 - **AI Mentorship Program Mentor:** Mentoring first and second-year undergraduate students about their future career, getting involved in AI research, and applying for industry or graduate school

- **Robotics Lunch Organizer (2019-2020):** Organizing bi-weekly robotics lunch sessions where invited professors, postdoctoral researchers and Ph.D. candidates present their research
- **Stanford STEM to SHTM Summer Internship Program, 2019** Stanford, CA, USA
 - **Mentor:** Mentoring high school students in a research project on decision making under risk / time constraints
- **IEEE Bilkent Student Branch, 2012-2017** Ankara, Türkiye
 - **Road to University Volunteer (2013-2017):** Introducing engineering and campus life to high school students from all around Türkiye
 - **Vice Chair (2014-2015):** Managing and organizing the technical trainings and social events
 - **Robotics and Automation Society Member (2012-2015):** Assistantship in the electronics and robotics focused tutorials
 - **Web Team Member (2012-2014) and Webmaster (2013-2014):** Designing and managing the website of IEEE Bilkent SB, its communities and events; teaching in web design tutorials and in MATLAB tutorials; organizing a Java programming competition and weekly brain teasers
- **GazeteBilkent, 2014-2017** Ankara, Türkiye
 - **Online Operations Manager:** Managing the website of GazeteBilkent, an online newspaper
- **Bilkent University, 2014-2017** Ankara, Türkiye
 - **Webmaster of Electrical and Electronics Engineering (2013-2017), Mechanical Engineering (2015-2017), Economics (2016-2017) Departments:** Designing, developing and managing the websites of the departments and developing necessary web tools
 - **1st and 2nd Industrial Design Projects Fairs Student Coordinator and Webmaster (2015, 2016):** Organizing the fair in which the senior students of Bilkent EEE present their industry projects
 - **Graduate Research Conference '15 Student Coordinator (2015):** Organizing the conference series in which graduate students of Bilkent EEE present their research and projects
 - **Graduate Research Conference '14 Webmaster and Organization Team Member (2014):** Organizing the conference series in which graduate students of Bilkent EEE present their research and projects
- **Bilkent Chess Society, 2013-2014** Ankara, Türkiye
 - **Vice Chair:** Organizing tournaments on campus, participating in local events as a team
- **Bilkent Academic Career Club, 2013** Ankara, Türkiye
 - **Physics Olympiads Volunteer:** Organizing a physics competition among high school students

PATENTS

1. Z Cao, **E Bıyık**, WZ Wang, A Raventos, A Gaidon, G Rosman, D Sadigh. “Reinforcement Learning Based Control of Imitative Policies for Autonomous Driving”, US Patent Application No. US17/002,650.

JOURNAL PUBLICATIONS

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